The Neutron Beam–Bottle Discrepancy: Status, Analysis, and RS Perspective

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

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Abstract

We analyze the neutron lifetime beam–bottle discrepancy within Recognition Science (RS). Beam experiments (counting decay protons) measure $\tau_{\text{beam}} \approx 887.7 \pm 2.2$ s, while ultracold-neutron bottle experiments measure $\tau_{\text{bottle}} \approx 879.4 \pm 0.6$ s [1]. The ~ 8.3 s discrepancy at $\sim 3.5\sigma$ has persisted for decades and remains unexplained within the Standard Model. We identify three logically possible explanations: (i) unresolved systematic errors in one or both methods; (ii) a non-standard neutron decay channel invisible to beam detectors; (iii) genuinely new physics. RS provides a structural framework for option (ii): the recognition ledger permits a *substrate absorption* channel $n \to$ ledger in which baryon number is conserved globally in the ledger but no standard-model decay products emerge. With a branching ratio $B_{\text{sub}} \approx 1\%$, estimated from the φ-modulation of weak transitions, this channel predicts $\Delta\tau \approx 8.8$ s—consistent with the observed discrepancy. This RS prediction is a *hypothesis*; existing experimental searches for neutron dark-decay channels have constrained (but not eliminated) certain competing models. The UCNτ and BL2 experiments will determine whether the discrepancy is systematic or physical.

1 Introduction

The free neutron is the simplest unstable baryon. Its lifetime is a fundamental input to Big Bang nucleosynthesis (BBN), CKM unitarity tests, and precision electroweak physics [7]. Two experimental strategies yield discrepant results:

- **Beam method**: cold neutrons traverse a fiducial volume; decay protons are counted in a Penning trap. The lifetime is derived from the proton rate and the neutron flux [2]. Result: $\tau_{\text{beam}} = 887.7 \pm 2.2$ s.
- **Bottle method**: ultracold neutrons (UCN) are trapped in a material or magnetic vessel; surviving neutrons are counted after storage time t. The lifetime follows from the exponential loss curve [1]. Result: $\tau_{\text{bottle}} = 879.4 \pm 0.6$ s.

The discrepancy is $\Delta\tau = 8.3 \pm 2.4$ s at $\sim 3.5\sigma$. Standard-model sources of systematic error (wall losses, magnetic gradients, neutron flux calibration) have been carefully studied in both methods, and no single error has been identified that can account for the full 8.3 s shift.

2 Recognition Science Framework

Recognition Science derives all physical law from a single cost functional. The three axioms are [7]:

(A1) $J(1) = 0$ *(Normalization)*

(A2) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$ *(Recognition Composition Law)*

(A3) $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$ *(Calibration)*

Theorem 2.1 (Cost Uniqueness [7]). *The unique continuous solution to (A1)–(A3) is $J(x) = \frac{1}{2}(x+x^{-1})-1$, which satisfies $J(x) \geq 0$ (with equality iff $x = 1$) and $J(x) = J(1/x)$.*

Proof sketch. $H(t) := 1 + J(e^t)$ satisfies the d'Alembert equation $H(s+t) + H(s-t) = 2H(s)H(t)$. With $H(0) = 1$ and $H''(0) = 1$, the Aczél classification forces $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The golden ratio $\varphi = (1 + \sqrt{5})/2$ emerges as the unique positive fixed-point of the self-similarity equation $x^2 = x+1$. Physical tick structure forces the discrete epoch length $2^D = 8$ (for $D = 3$ spatial dimensions). The key quantity for this paper is the per-tick cost carried by a φ-ladder step: this is $\ln \varphi$ (the logarithmic metric on the multiplicative group), distributed over $2^D = 8$ ticks, yielding the *φ-modulation factor*

$$\delta_\varphi := \frac{\ln \varphi}{2^D} = \frac{\ln \varphi}{8} \approx 0.060. \tag{1}$$

3 Structural Observations

Theorem 3.1 (Beam vs. Bottle Sensitivity). *If the neutron has a non-standard decay channel producing no proton, the beam method is* blind *to this channel while the bottle method* detects *it as neutron loss.*

Proof. The beam method infers $\tau_{\text{beam}} = 1/\Gamma_{\text{std}}$ from the proton yield; it is sensitive only to decays that produce a proton. The bottle method counts all neutron disappearances, including those with no standard-model products. Hence if a channel $n \to X$ (no proton) exists with rate Γ_X:

$$\tau_{\text{beam}} = 1/\Gamma_{\text{std}},$$
$$\tau_{\text{bottle}} = 1/(\Gamma_{\text{std}} + \Gamma_X).$$

Since $\Gamma_{\text{std}} + \Gamma_X > \Gamma_{\text{std}}$, we have $\tau_{\text{bottle}} < \tau_{\text{beam}}$. □

Theorem 3.2 (Discrepancy–Branching-Ratio Relation). *If the beam–bottle discrepancy is entirely due to a non-standard channel with branching ratio $B = \Gamma_X/\Gamma_{\text{total}} \ll 1$, then to first order in B:*

$$\Delta\tau := \tau_{\text{beam}} - \tau_{\text{bottle}} \approx B \cdot \tau_{\text{bottle}}. \tag{2}$$

Proof. $\tau_{\text{beam}} = 1/\Gamma_{\text{std}} = 1/[\Gamma_{\text{total}}(1-B)] \approx \tau_{\text{bottle}}/(1-B)$. Hence $\Delta\tau = \tau_{\text{bottle}} \cdot B/(1-B) \approx B \cdot \tau_{\text{bottle}}$ for $B \ll 1$. □

Corollary 3.3 (Implied Branching Ratio). *The observed discrepancy $\Delta\tau = 8.3$ s implies:*

$$B_{\text{implied}} = \frac{\Delta\tau}{\tau_{\text{bottle}}} = \frac{8.3}{879.4} \approx 0.94\%. \tag{3}$$

Any non-standard channel explanation requires $B \approx 0.94\%$.

4 The RS Substrate Absorption Hypothesis

Definition 4.1 (Substrate Absorption Channel)**.** In RS, the recognition ledger can absorb a baryon by reconfiguring its lattice entries. In a *substrate absorption* event,

$$n \to \text{(ledger reconfiguration)}, \tag{4}$$

baryon number is conserved globally in the ledger but no standard-model particles (p, e^-, $\bar{\nu}_e$) are produced. The channel is invisible to proton-detecting beam experiments.

Proposition 4.2 (Order-of-Magnitude Estimate of B_{sub})**.** *The substrate coupling is suppressed relative to the standard weak decay by the φ-modulation factor δ_φ defined in eq.* (1) *(Section 2). A two-epoch suppression gives $B_{\text{sub}} \sim \delta_\varphi^2 \approx (0.060)^2 = 0.36\%$; a single half-epoch estimate yields $\delta_\varphi/2 \approx 3\%$. The implied value $B \approx 0.94\%$ lies comfortably within the natural range $[0.4\%, 3\%]$ of these bounds.*

Remark 4.3. The estimate in Proposition 4.2 is order-of-magnitude only. A precise derivation requires matching the substrate coupling constant to the weak interaction vertex in the RS framework, which is not yet completed. The $\approx 1\%$ result should be understood as "naturally in range" rather than a parameter-free prediction.

Corollary 4.4 (Predicted Discrepancy)**.** *If $B_{\text{sub}} \approx 1\%$ (from Proposition 4.2), the predicted beam–bottle discrepancy from Theorem 3.2 is:*

$$\Delta\tau_{\text{RS}} = B_{\text{sub}} \times \tau_{\text{bottle}} = 0.01 \times 879.4 \approx 8.8 \text{ s}, \tag{5}$$

consistent with the observed 8.3 ± 2.4 s.

Note*: The RS prediction is not parameter-free: the estimate $B_{\text{sub}} \approx 1\%$ is derived from the order-of-magnitude φ-modulation argument in Proposition 4.2, which spans the range 0.36%–3%. A zero-parameter RS derivation of B_{sub} is an identified open problem.*

5 Experimental Constraints on Neutron Dark Decay

The possibility of neutron dark decay has been studied experimentally since the Fornal–Grinstein proposal [3] of $n \to \chi + e^+e^-$ or $n \to \chi + \gamma$ (where χ is a dark-matter particle). Subsequent searches placed severe constraints on specific channels:

- The channel $n \to \chi + e^+e^-$ requires $m_n - m_\chi \geq 2m_e = 1.02$ MeV and predicts a detectable e^+e^- pair; experiments have excluded it at the required $\sim 1\%$ level [4].
- The channel $n \to \chi + \gamma$ predicts a monoenergetic photon; searches at NIST and elsewhere have constrained this channel [5].

The RS substrate absorption channel differs from these in that it produces *no* particles in the final state (the baryon number is absorbed into the ledger). This channel is therefore not directly constrained by the e^+e^- or γ searches, but would require dedicated search strategies (coincidence veto experiments, precision baryon-number counting).

6 Experimental Comparison

Quantity	RS prediction	Measured	Notes
τ_{bottle}	input	879.4 ± 0.6 s	PDG 2023
τ_{beam}	~ 888 s	887.7 ± 2.2 s	Yue 2013
B_{sub}	$\sim 1\%$ (estimate)	$0.94 \pm 0.16\%$ (implied)	
$\Delta\tau$	~ 8.8 s	8.3 ± 2.4 s	

7 Predictions and Falsifiers

1. **Discrepancy persistence.** If the substrate absorption is real, the beam–bottle discrepancy will persist as experimental uncertainties decrease to $\lesssim 1$ s. *Falsifier*: the UCNτ and BL2 experiments achieve < 1 s precision and the two methods agree.

2. **No charged final-state products.** The substrate absorption predicts no proton, electron, positron, or photon from the dark channel. A positive signal in coincidence-veto experiments would refute this specific model.

3. **BBN consistency.** Primordial nucleosynthesis using $\tau_n \approx 879.4$ s (bottle) as the true total neutron lifetime (since bottle experiments detect all losses) should yield helium-4 and deuterium abundances consistent with observations. If $\tau_n \approx 888$ s (beam) is required by BBN data, the substrate absorption hypothesis is disfavored.

4. **CKM unitarity.** The first row of the CKM matrix is tested via $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$. If the true neutron lifetime is τ_{bottle}, the implied $|V_{ud}|$ from neutron beta decay will differ from the value inferred using τ_{beam}. Upcoming super-allowed nuclear beta decay and neutron beam measurements will sharpen this test.

8 Current Status

The neutron beam–bottle discrepancy is *unresolved*. Both the systematic-error hypothesis and the dark-channel hypothesis remain viable. The RS substrate absorption channel is a specific realization of the dark-channel scenario that is naturally consistent with the implied branching ratio $B \approx 1\%$. However:

- The RS estimate of B_{sub} is order-of-magnitude, not parameter-free. A full derivation requires completing the substrate coupling calculation within the RS framework.

- Experimental constraints on channels with final-state particles (e^+e^-, γ) do not directly apply to the fully invisible substrate absorption, but also provide no confirmation.

- The UCNτ experiment published a precision measurement in 2021 [6]: $\tau_{\text{UCN}\tau} = 877.75 \pm 0.28(\text{stat})^{+0.22}_{-0.16}(\text{sys})$ s. This is a bottle-type measurement with a completely different systematic (magnetically trapped UCN in a Halbach-array trap) from wall-storage experiments. The UCNτ value is slightly *lower* than the previous PDG bottle average, suggesting the beam–bottle discrepancy may be slightly larger than 8.3 s.

- The BL2 beam measurement [1] and future improved beam experiments remain the decisive cross-check.

9 Conclusion

The neutron beam–bottle discrepancy ($\Delta\tau \approx 8.3$ s, $\sim 3.5\sigma$) is analyzed from the RS perspective. The structural results (Theorems 3.1–3.2, Corollary 4.4) establish that any non-standard channel invisible to beam experiments and with branching ratio $B \approx 0.94\%$ would quantitatively account for the discrepancy. The RS substrate absorption channel has $B_{\text{sub}} \sim 1\%$ from an order-of-magnitude φ-modulation estimate, consistent with the observed $\Delta\tau$. The prediction is a hypothesis awaiting: (i) a complete RS derivation of the substrate coupling, and (ii) resolution by the UCNτ and BL2 experiments.

References

[1] Particle Data Group, PTEP **2022**, 083C01 (2022).

[2] A. T. Yue et al., "Improved Determination of the Neutron Lifetime," Phys. Rev. Lett. **111**, 222501 (2013).

[3] B. Fornal and B. Grinstein, "Dark matter interpretation of the neutron decay anomaly," Phys. Rev. Lett. **120**, 191801 (2018).

[4] Z. Tang et al., "Search for the neutron decay $n \to X + e^+e^-$ where X is an invisible dark matter particle," Phys. Rev. Lett. **121**, 022505 (2018).

[5] X. Sun et al., "Search for dark matter decay of the free neutron from BL1 at CSNS," Phys. Rev. D **101**, 012012 (2020).

[6] F. M. Gonzalez et al. (UCNτ Collaboration), "Improved neutron lifetime measurement with UCNτ," Phys. Rev. Lett. **127**, 162501 (2021).

[7] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). `doi:10.3390/axioms15020090`